

SULFURIC ACID

MATERIAL SAFETY DATA SHEET

Distributed By: PVS NOLWOOD CHEMICALS, INC. 10900 Harper Avenue Detroit, MI. 48213 (313) 925-0300	
PVS ITEM #	PVS MSDS #
Multi	0738

CONTROLLED DOCUMENT
IF STAMPED IN RED

Section 1. Chemical Product and Company Identification

Product Name: Sulfuric Acid
Synonym: Sulphuric Acid, Oil of Vitriol
Chemical Name: Sulfuric Acid
Chemical Family: Inorganic Mineral Acid
Chemical Formula: H_2SO_4
C.A.S. Number: 7664-93-9
Manufacturer: PVS Chemical Solutions
55 Lee Street
Buffalo, NY 14210
Emergency Contact: CHEMTREC 1-800-424-9300
Technical Contact: 716/825-5762
Revision Date: November 1, 2010

Section 2. Composition, Information on Ingredients

<u>Component(s)</u>	<u>Percent</u>	<u>CAS Reg No.</u>	<u>Hazardous</u>
Sulfuric Acid	75-100% (typ)	7664-93-9	Yes
Water	0-25% (typ)	7732-18-5	No

Section 3. Health and Hazard Information

EMERGENCY OVERVIEW: Causes eye and skin burns. Liquid and mist cause severe burns to all body tissue. Causes digestive and respiratory tract burns. May be fatal if swallowed or if mist is inhaled. Strong inorganic acid mists containing sulfuric acid may cause cancer. Concentrated sulfuric acid reacts violently with water and other substances under certain conditions.

POTENTIAL HEALTH EFFECTS:

Eye Contact: Contact with eyes may result in permanent visual loss unless removed quickly by thorough irrigation with water.

Inhalation: Inhalation of concentrated vapor or mist may damage respiratory tract. Inhalation may be fatal. May cause irritation to the respiratory tract with burning pain in the nose and throat, coughing, wheezing, shortness of breath and pulmonary edema.

Ingestion: Swallowing may be fatal. Causes severe and permanent damage to the digestive tract

Skin Contact: Contact with liquid, mist, or vapor can cause immediate irritation or corrosive burns to all human tissue. Severity of the burn is generally determined by the concentration of the solution and duration of exposure.

Skin Absorption:

Absorption is unlikely to occur.

Chronic Effect:

Prolonged or repeated skin contact may cause dermatitis. Prolonged or repeated exposure may cause erosion of teeth, perforation of nasal septum or bronchitis. Exposure to strong inorganic acid mists containing sulfuric acid is carcinogenic to humans.

Section 4. First Aid Measures

Inhalation:

Remove victim to fresh air. If not breathing, perform artificial respiration. If breathing is difficult, give oxygen. Get immediate medical attention.

Ingestion:

Do not induce vomiting. Drink copious amounts of water or milk. Get immediate medical attention. Never give anything by mouth to an unconscious person.

Direct Contact:

Wipe off excess. Flush immediately with water for at least 15 minutes while removing contaminated clothing. Get immediate medical attention. Wash clothing before re-use. Destroy contaminated shoes.

Direct Eye Contact:

Flush immediately with water for at least 15 minutes. Forcibly hold eyelids apart to ensure complete irrigation of eye/lid tissue. Get immediate medical attention.

Section 5. Fire Fighting Measures

Flash Point:

Not flammable. May ignite combustible materials.

Flammable Limits In Air:

Lower: not applicable Upper: not applicable

Extinguishing Media:

Use dry chemical or CO₂ fire extinguishers to fight surrounding fire. Do not use water on acid itself. Apply from farthest possible distance.

Special Fire Fighting Procedures: Wear self-contained breathing apparatus and full protective clothing. Cool exterior of storage tanks.

Unusual Fire And Explosion Hazards: Violent reaction with water. Evolution of explosive hydrogen gas on contact with most metals. Will react with organic material with evolution of heat and dense white fumes.

Section 6. Accidental Release Measures

Spill or Leak:

Utilize full protective clothing, including boots and protective equipment. Contain spill in order to prevent contamination of sewage system or waterway. Pump into marked containers for reclamation or disposal. If possible, neutralize on a dry basis with suitable alkali such as lime or soda ash; then flush with

water in accordance with applicable regulations. Do not get water inside containers.

Section 7. Handling & Storage

Storage: Store in a cool dry well ventilated area. Do not store near combustible materials, alkaline substances or other incompatible materials. Store protected from moisture. Keep container closed when not in use. Protect container from physical damage. Diking of storage tanks is recommended.

Normal Handling: Wash thoroughly after handling. Remove contaminated clothing and wash before reuse. Do not allow water to enter container due to violent reaction. Use only with adequate ventilation. When diluting, always add the acid to the water to avoid dangerous spattering.

Section 8. Exposure Controls and Personal Protection

Exposure Limits: OSHA PEL: 1 mg/m³ (TWA)
ACGIH TLV: 0.2 mg/m³ (TWA)
NIOSH: 1mg/m³, 15 mg/m³ IDLH

Ventilation: Provide ventilation to control exposure levels below airborne exposure limits. Use local exhaust ventilation. Reference NFPA Standard 91 for design of exhaust systems.

Respiratory Protection: When exposure limit is exceeded and engineering controls are not feasible, use NIOSH/MSHA approved, full face respirator with cartridges approved for sulfuric acid vapor and mist. Consult respirator manufacturer to determine appropriate equipment. If concentrations are high or unknown, use self-contained breathing apparatus.

Protective Gloves: Wear impervious rubber gloves.

Eye Protection: Wear splash proof chemical safety goggles. Eyewash fountains recommended in all storage and handling areas. Do not wear contact lenses.

Other Protective Equipment: Wear protective clothing to prevent skin contact. Full face shield and rubber footwear should be used. Acid-resistant hood and full body suit recommended. Safety shower recommended in all storage and handling areas.

Work/Hygienic Practices: Avoid breathing fumes. Use gloves when handling. Remove and change contaminated clothing immediately.

Section 9. Physical and Chemical Properties

Appearance And Odor: Clear to slightly cloudy, oily liquid; Odorless to slightly pungent.

Molecular Weight: 98.08

Boiling Point: 77.7% = 193⁰C; 93% = 279⁰C; 96% = 308⁰C; 98% = 327⁰C;
99% = 310⁰C

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Freezing Point: 77.7% = -11.4 °C; 93% = -29 °C; 96% = -14 °C; 98% = -1 °C; 99% = 4.4 °C
Vapor Pressure (Reid): 93.2% = 0.0016 mm Hg; 98% = 0.002 mm Hg
Vapor Density (Air = 1): 3.4
Specific Gravity (Water = 1): 77.7% = 1.706; 93.2% = 1.835; 96% = 1.843; 98% = 1.844; 99% = 1.842
Percent Volatile By Weight: no information
Evaporation Rate (Butyl Acetate = 1): <1
pH: <1
Solubility In Water: Completely miscible, liberates heat

Section 10. Stability and Reactivity

Stability: Stable under ordinary conditions and use.
Hazardous Polymerization: Will not occur.
Incompatibility (Conditions And Materials To Avoid): Material is stable when properly handled. Concentrated acid reacts violently with water, spattering and liberating heat. Reactive with materials such as metals, metal oxides, hydroxides, nitrates, amines, carbonates and other alkaline materials. Reacts with cyanides and sulfides to form poisonous hydrogen cyanide and hydrogen sulfide respectively. Reacts with carbonates to generate carbon dioxide gas. Incompatible with bases, halogens, nitrates, permanganates, chlorates, perchlorates, metal acetylides, fulminates, picrates and organic materials. Reactions can generate a great deal of heat as does the dilution of acid with water. Concentrated acid is a strong oxidizing agent. May cause ignition of combustible materials on contact with generation of sulfur dioxide fumes. Avoid excessive heat, open flames or sparks.
Hazardous Decomposition Products: Explosive hydrogen gas is generated by the action of acid on most metals and may accumulate in metal containers. Releases Sulfur Dioxide at extremely high temperatures.

Section 11. Toxicological Information

Toxicity Data (Animal): Oral, rat: LD₅₀ = 2140 mg/kg
Inhalation, rat: LC₅₀ = 510 mg/m³
Standard Draize, eye rabbit 250 ug (severe)
Skin and eye irritation (rabbit): (FHSA) Corrosive
Carcinogenicity: The International Agency of Research on Cancer (IARC) has classified "strong inorganic acid mists containing sulfuric acid" as a Category 1 carcinogen, a substance that is "carcinogenic to humans". This classification is for inorganic acid mists only and does not apply to sulfuric acid or sulfuric acid solutions.
Medical Conditions Aggravated By Exposure: Repeated skin contact with dilute solutions may cause dermatitis. May cause dental erosion.

Section 12. Ecological Information

Ecotoxic Effects: Harmful to aquatic life in very low concentrations. May be dangerous if it enters water intake; Fish toxicity; 2.8 µg/L 96 hrs LC50 Rainbow trout.

Section 13. Disposal Considerations

Waste Disposal Methods: Dispose of spilled, neutralized or waste product, contaminated soil and other materials in accordance with all local, state and federal regulations. Waste sulfuric acid would exhibit the RCRA characteristic of corrosivity and should be handled as a Hazardous Waste. Do not flush to surface water or sanitary sewer.

Section 14. Transportation Information

USDOT:

Proper shipping name: Sulfuric acid (with more than 51% acid)
Hazard Class: 8
UN Number: UN1830
Packing Group: II
Reportable Quantity: 1000 pounds (454 kg)
IMO: 8
IATA/ICAO Class: 8
ERG Guide: 137

Canada (Under TDG)

Shipping Name: Sulfuric acid (with more than 51% acid)
Classification(s): Class 8
Product Identification No: UN1830
Packing Group: II
Regulated Limit: 50 kg

Section 15. Regulatory Information

SARA Title III: Sulfuric Acid (aerosol forms only) is a toxic chemical subject to the reporting requirements of Section 313 of Title III of the Superfund Amendments and Reauthorization Act (SARA) of 1986 and 40 CFR Part 372.

SARA Title III Sec 302: Listed as an Extremely Hazardous Substance with a TPQ = 1000lbs.

SARA Title III Sec 312: Acute, Chronic, Reactivity

SARA Title III Sec 313: Listed as "aerosol form only".

RCRA 261.33: No
TSCA 8(d): Listed

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CERCLA: RQ = 1000 lbs.
WHMIS Classification: Class E – Corrosive, Class D1A – Very Toxic, Class D2B – Suspected Human Carcinogen.
WHMIS Health Effects Index: Acute Lethality - very toxic – immediate
Materials Causing Other Toxic Effects- Chronic
Corrosive to animal skin

Other Ratings:

(hazard index key: 4=severe, 3=serious, 2=moderate, 1=slight, 0=minimal)

HMIS: Health=3, Flammability=0, Reactivity=2

NFPA: Health=3, Flammability=0, Reactivity=2

Section 16. Other Information

For additional Non-emergency health, safety, or environmental information, contact:

PVS Chemical Solutions
55 Lee Street
Buffalo, New York 14210
(716) 825-5762

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